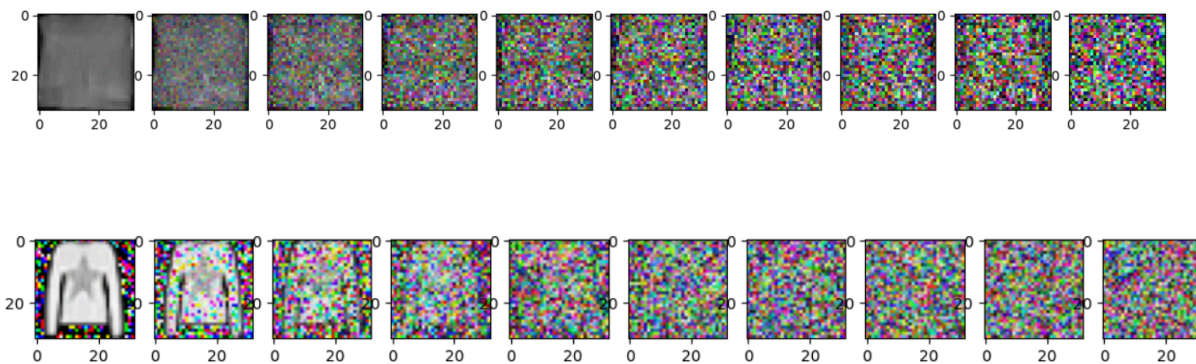


Diffusion Model

The Diffusion model notebook with 20 epochs produces the following result on Fashion-MNIST:

- Loss: 0.0692

```
Starting Epoch 20/20
Epoch 20 | Step 0 | Loss: 0.0784
Epoch 20 | Step 100 | Loss: 0.0922
Epoch 20 | Step 200 | Loss: 0.0819
Epoch 20 | Step 300 | Loss: 0.0696
Epoch 20 | Step 400 | Loss: 0.0742
Epoch 20 | Step 500 | Loss: 0.0796
Epoch 20 | Step 600 | Loss: 0.0843
Epoch 20 | Step 700 | Loss: 0.0796
Epoch 20 | Step 800 | Loss: 0.0861
Epoch 20 | Step 900 | Loss: 0.0662
Epoch 20 | Step 1000 | Loss: 0.0692
```

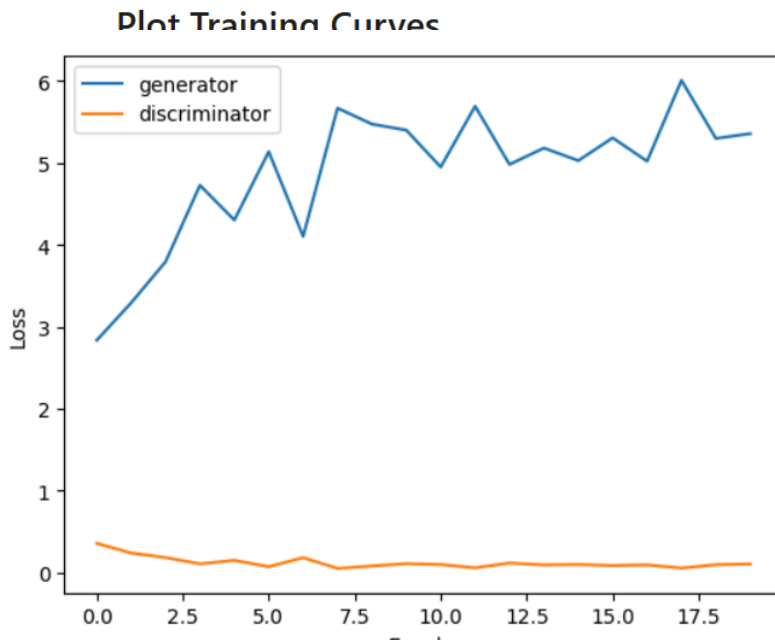


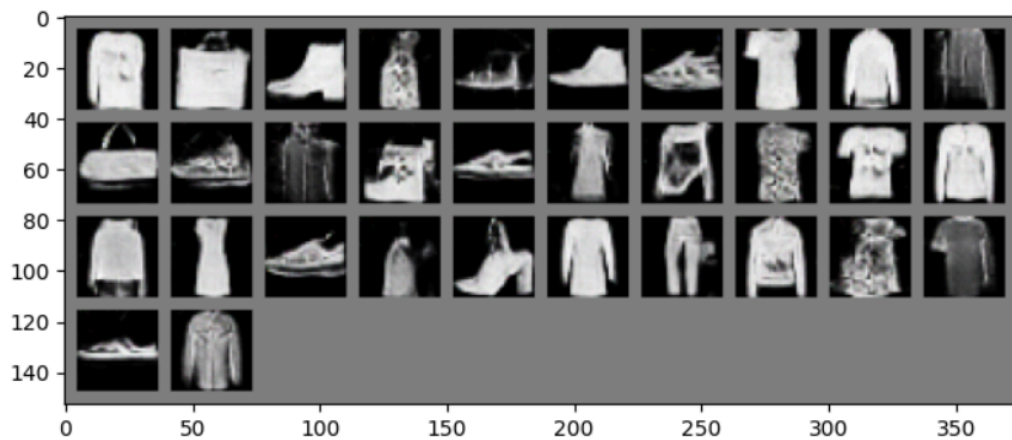
GAN

The GAN notebook with 20 epochs produces the following result on Fashion-MNIST:

- Generator Loss: **5.35**
- Discriminator Loss: **0.100**

```
Epoch [1 / 20] average loss generator vs. discrim.: 3.792951 vs. 0.179036
Epoch [2 / 20] average loss generator vs. discrim.: 4.727635 vs. 0.102205
Epoch [3 / 20] average loss generator vs. discrim.: 4.302822 vs. 0.145116
Epoch [4 / 20] average loss generator vs. discrim.: 5.137404 vs. 0.067655
Epoch [5 / 20] average loss generator vs. discrim.: 4.103621 vs. 0.177412
Epoch [6 / 20] average loss generator vs. discrim.: 5.670535 vs. 0.046362
Epoch [7 / 20] average loss generator vs. discrim.: 5.475391 vs. 0.075869
Epoch [8 / 20] average loss generator vs. discrim.: 5.401425 vs. 0.104170
Epoch [9 / 20] average loss generator vs. discrim.: 4.950779 vs. 0.093627
Epoch [10 / 20] average loss generator vs. discrim.: 5.693490 vs. 0.053360
Epoch [11 / 20] average loss generator vs. discrim.: 4.982023 vs. 0.111688
Epoch [12 / 20] average loss generator vs. discrim.: 5.182846 vs. 0.088993
Epoch [13 / 20] average loss generator vs. discrim.: 5.028824 vs. 0.093947
Epoch [14 / 20] average loss generator vs. discrim.: 5.307380 vs. 0.081699
Epoch [15 / 20] average loss generator vs. discrim.: 5.021874 vs. 0.089371
Epoch [16 / 20] average loss generator vs. discrim.: 6.010028 vs. 0.050182
Epoch [17 / 20] average loss generator vs. discrim.: 5.299878 vs. 0.091841
Epoch [18 / 20] average loss generator vs. discrim.: 5.358650 vs. 0.100473
```





Comparison

- GAN generates images faster because the Generator creates samples in a single pass
- Diffusion models generate images through gradual denoising, making training and sampling slower than GANs
- In this analysis, the GAN model produced sharper and more recognizable clothing shapes, while the diffusion model generated smoother outputs but required slower training and image generation.